Money in Flexible Price Environments: CIA

Cash-in-Advance Model

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Readings

- Walsh, C. E., 2010. Monetary Theory and Policy, 3rd Ed. The MIT Press. (CH 3)
- Cooley, T. F. and G. D. Hansen, 1989. The inflation tax in a real business cycle model. *American Economic Review*, 79: 733-748.

Introduction

- The Neoclassical growth model is a model of nonmonetary economy, in which goods are exchanged and transactions must be taking place. In the economy, there is neither medium of exchange (money) that is used to facilitate these transactions, nor an asset like money that has a zero nominal return and is therefore dominated in rate of return by other interest-bearing assets.
- In order to study monetary issues using the Neoclassical framework, we need to introduce money into the model economy and motivate agents to hold positive quantity of money
- There are different approaches to introducing money into the Neoclassical growth model. We will study the most popular two, the cash in advance (CIA) model and money in utility (MIU) model.

Introduction

- The MIU model (Sidrauski, 1967) assumes that money itself yield direct utility by incorporating real money balances in the utility functions of the agents of the model, whereas in the CIA model, money is a medium of exchange for purchasing consumption goods.
- In this session, we will look at the CIA model due to Clower (1967) and Lucas (1982). The model described here is the Cooley-Hansen(1989) CIA model.

The Model

Firm

Aggregate output is produced according to the following constant returns to scale technology:

$$y_t = f(k_{t-1}, n_t, z_t) = e^{z_t} k_{t-1}^{\alpha} n_t^{1-\alpha} \tag{1}$$

where the technology shock z_t follows:

$$z_t = \rho_z z_{t-1} + \epsilon_{z,t}, \qquad 0 < \rho_z < 1, \qquad \epsilon_{z,t} \sim i.i.d.(0, \sigma_z^2) \tag{2}$$

and the capital accumulation equation:

$$k_t = (1 - \delta) k_{t-1} + x_t \tag{3}$$

The Model

Household

Household maximizes the discounted expected utility function:

$$E_0 \sum_{t=0}^{\infty} \beta^t U(c_t, 1 - n_t) = E_0 \sum_{t=0}^{\infty} \beta^t [\frac{c_t^{1-\eta_c}}{1-\eta_c} + \frac{(1-n_t)^{1-\eta_n}}{1-\eta_n}] \quad (4)$$

s.t.
the budget constraint:

$$y_t + \tau_t + (1-\delta)k_{t-1} + \frac{m_{t-1}}{1+\pi_t} + \frac{b_{t-1}(1+i_{t-1})}{1+\pi_t} \geq c_t + k_t + m_t + b_t \quad (5)$$

and the CIA constraint:

$$C_t \leq M_{t-1} + T_t - B_t, \quad or \quad c_t \leq \frac{m_{t-1}}{1+\pi_t} + \tau_t - b_t \quad (in\ real\ term) \quad (6)$$

The Model

Government budget constraint

$$T_t = \bar{M}_t - \bar{M}_{t-1} \tag{7}$$

Money creation process

- Money growth

$$\bar{M}_t = (1 + \theta_t)\bar{M}_{t-1} \tag{8}$$

- Money supply shock

$$\mu_t = \rho_m \mu_{t-1} + \phi z_{t-1} + \epsilon_{m,t} \qquad \rho_m > 0, \qquad \epsilon_{m,t} \sim i.i.d.(0, \sigma_m^2) \tag{9}$$

where

$$\mu_t = \theta_t - \bar{\theta} \tag{10}$$

The Model

Remarks

- Timing: The timing is that assets market opens first (Lucas, 1982). Agents only hold an amount of money that is just sufficient to finance their desired level of consumption at the beginning of each period, *after observing any current shocks*. In other words, household is able to adjust its portfolio between money and other assets (bonds) before entering the goods market (Walsh: 101).
- In the case of goods market opens first (Svensson, 1985), households enter time period t with money holding m_{t-1} and receives a lump-sum transfer τ_t. Households cannot adjust their portfolio after shocks hit[1].
- Since the opportunity cost of holding money is positive as long as the nominal interest rate is greater than zero, ρ_m should always be greater than zero. The CIA constraint (6) is always binding.

[1] Since we focus on the impact of money, without loss of generality, we exclude bonds in the model.

Solving the Model

Household's decision is characterized by the following value function:

$$V(k_{t-1}, m_{t-1}, z_t, \mu_t) = \max_{c_t, n_t, k_t, m_t} [U(c_t, 1 - n_t) + \beta E_t V(k_t, m_t)] \quad (11)$$

where

$$U(c_t, 1 - n_t) = \frac{c_t^{1-\eta_c}}{1 - \eta_c} + \frac{(1 - n_t)^{1-\eta_n}}{1 - \eta_n} \quad ((4))$$

Following Walsh (2003)[2]:

- define $\omega_t = y_t + (1 - \delta)k_{t-1} + \tau_t + \frac{m_{t-1}}{1+\pi_t}$
- define λ_t and Λ_t are the multipliers of budget constraint and CIA constraint respectively

[2] $y_t = f(k_{t-1}, n_t, z_t) = z_t k_{t-1}^{\alpha} n_t^{1-\alpha}$

Solving the Model

FOCs w.r.t. $\{c_t, n_t, k_t, m_t\}$:

$$\lambda_t + \Lambda_t = U_c(c_t, 1 - n_t) \tag{12}$$

$$(1-\alpha) z_t k_{t-1}^{\alpha} n_t^{-\alpha} \lambda_t = U_n(c_t, 1 - n_t) \tag{13}$$

$$\beta R_t E_t V_k(k_t, m_t) = \lambda_t \tag{14}$$

$$\beta E_t\left[\frac{\lambda_{t+1} + \Lambda_{t+1}}{1 + \pi_{t+1}}\right] = \lambda_t \tag{15}$$

Solving the Model

Remarks

☐ Recall (12):

$$\lambda_t + \Lambda_t = U_c(c_t, 1 - n_t) \qquad ((12))$$

MU_c is the sum of the MU_ω, λ_t, and the value of the liquidity service (cost of holding money), Λ_t, needed to finance the transaction.

☐ Substitute $\lambda_{t+1} = V_k(k_t, m_t)$ into (14):

$$\beta R_t E_t \lambda_{t+1} = \lambda_t \qquad (16)$$

(16) is the standard asset pricing equation. The MC from reducing wealth today, λ_t, must equal to the utility value of carrying that wealth forward one period, earning a gross real rate of return R_t, where tomorrow's utility is discounted back to today at rate β (Walsh 2003: 105).

Solving the Model

Remarks

□ Recall (15):

$$\beta E_t[\frac{\lambda_{t+1} + \Lambda_{t+1}}{1 + \pi_{t+1}}] = \lambda_t \qquad ((15))$$

$$\beta E_t[\frac{\lambda_{t+1} + \Lambda_{t+1}}{P_{t+1}/P_t}] = \lambda_t \qquad (17)$$

$$\beta E_t[\frac{\lambda_{t+1} + \Lambda_{t+1}}{P_{t+1}\lambda_t}] = \frac{1}{P_t} \qquad (18)$$

(18) is the asset pricing equation for money. The price/value of a unit of money in terms of goods, $\frac{1}{P_t}$, is equal to discounted marginal benefit from future earnings (value of money).

Why Inflation Tax?

In Cooley-Hansen CIA model, a positive nominal interest rate acts as a tax on consumption. Why?

Combining (15) and (16), we have:

$$\beta E_t[\frac{\lambda_{t+1} + \Lambda_{t+1}}{1 + \pi_{t+1}}] = \beta R_t E_t \lambda_{t+1} \tag{19}$$

or

$$E_t[\lambda_{t+1} + \Lambda_{t+1}] = R_t E_t[\lambda_{t+1}(1 + \pi_{t+1})] \tag{20}$$

Since $1 + i_t = R_t(1 + \pi_{t+1})$, (19) becomes[3]:

$$1 + i_t = (\frac{\lambda_{t+1} + \Lambda_{t+1}}{\lambda_{t+1}}) \tag{21}$$

[3] expectation is dropped here.

Why Inflation Tax?

$$i_t = \left(\frac{\lambda_{t+1} + \Lambda_{t+1}}{\lambda_{t+1}}\right) - 1 \tag{22}$$

$$i_t = \frac{\Lambda_{t+1}}{\lambda_{t+1}} \tag{23}$$

That is, if the CIA constraint is binding, then money has value because it yields valued liquidity services. In this case, the nominal interest rate is positive and it acts as a tax on consumption.

Inflation represents a tax on the purchases of the cash goods, and therefore higher rate of inflation shifts household demand away from cash good (consumption) and toward the credit good (such as leisure and investment).

Steady States

$$R^{ss} = \frac{1}{\beta} \tag{24}$$

$$\frac{y^{ss}}{k^{ss}} = \frac{1}{\alpha}(R^{ss} - 1 + \delta) \tag{25}$$

$$\frac{c^{ss}}{k^{ss}} = \frac{y^{ss}}{k^{ss}} - \delta \tag{26}$$

$$\frac{n^{ss}}{k^{ss}} = \left(\frac{y^{ss}}{k^{ss}}\right)^{\frac{1}{1-\alpha}} \tag{27}$$

$$\frac{(n^{ss})^{\eta_c}}{(1-n^{ss})^{\eta_n}} = (1-\alpha)\frac{\beta}{1+\pi^{ss}}\left(\frac{c^{ss}}{k^{ss}}\right)^{-\eta_c}\left(\frac{y^{ss}}{k^{ss}}\right)^{\frac{\eta_c-\alpha}{1-\alpha}} \tag{28}$$

Steady States

□ Welfare cost of inflation in CIA model (Cooley-Hansen model)

In steady state, the binding CIA constraint gives:

$$c^{ss} = \frac{m^{ss}}{1+\pi^{ss}} + \tau^{ss} = m^{ss} \tag{29}$$

Therefore:

$$\frac{c^{ss}}{k^{ss}} = \frac{m^{ss}}{k^{ss}} \tag{30}$$

Recall (28):

$$(1-n^{ss})^{-\eta_n}(n^{ss})^{\eta_c} = (1-\alpha)\frac{\beta}{1+\pi^{ss}}\left(\frac{c^{ss}}{k^{ss}}\right)^{-\eta_c}\left(\frac{y^{ss}}{k^{ss}}\right)^{\frac{\eta_c-\alpha}{1-\alpha}} \tag{(28)}$$

$\Rightarrow n^{ss} \downarrow$ if $\theta^{ss} \uparrow$ (with the CIA constraint binding, in steady state $\pi^{ss} = \theta^{ss}$)

Steady States

☐ The superneutrality of money

- The model exhibits the neutrality property: proportional changes in the level of nominal money balances and prices do not affect real variables.
- In (24) $\backsim$ (27), none of the steady state values/ratios depend on the growth rate of nominal money supply — CIA model exhibits superneutrality iff labor supply is inelastic (fixed hours, $n = \bar{n}$). With endogenous labour and fixed preferences, (28) implies that trend inflation changes steady-state hours.

☐ The optimal rate of inflation

An efficient equilibrium will be characterized by equality between social and private costs. Because the social cost of producing money is zero, the private opportunity cost of holding money must be zero in order to achieve optimality.

The private opportunity cost is measured by the nominal interest rate, so the optimal rate of inflation in the steady state is the rate that achieves a zero nominal rate of interest — deflation (Walsh 2003: 126).

CIA: interpreting the code and Walsh

- Walsh (2010), Appendix 3.6, pp. 128–130, uses utility over consumption and leisure, not power disutility of hours. Thus the code's $1 + \eta_n n^{ss}/(1 - n^{ss})$ is consistent with leisure curvature. The utility comment in `CIA.mod` describes different preferences.
- λ (`lambdda`) is the marginal value of wealth; with binding cash constraints it differs from marginal utility of consumption. The two multipliers in the nonlinear derivation must not be conflated.
- The code uses $i_t = r_t + E_t\pi_{t+1}$ for rate deviations, a small-rate approximation. Exact Fisher scaling gives $i_t/(1 + i^{ss}) = r_t/(1 + r^{ss}) + E_t\pi_{t+1}/(1 + \pi^{ss})$. Walsh's collected CIA equations retain $1/(1 + i^{ss})$ in Eq. 3.73.
- Forward leads in Dynare represent conditional expectations here. The code retains the direct return-on-capital equation; Walsh eliminates expected output to obtain his equivalent reduced representation, subject to the stated rate approximations.
- `nss` is calibrated. Inflation comparisons at fixed `nss` implicitly recalibrate the leisure weight, rather than tracing steady states under fixed preferences.

Dynamics

The linearized model:

$$\hat{y}_t = \alpha\hat{k}_{t-1} + (1-\alpha)\hat{n}_t + z_t \tag{31}$$

$$(\frac{y^{ss}}{k^{ss}})\hat{y}_t = (\frac{c^{ss}}{k^{ss}})\hat{m}_t + \hat{k}_t - (1-\delta)\hat{k}_{t-1} \tag{32}$$

$$\hat{r}_t = \alpha(\frac{y^{ss}}{k^{ss}})(E_t\hat{y}_{t+1} - \hat{k}_t) \tag{33}$$

$$\hat{\lambda}_t = E_t\hat{\lambda}_{t+1} + \hat{r}_t \tag{34}$$

$$\hat{y}_t + \hat{\lambda}_t = (1 + \eta_n\frac{n^{ss}}{1-n^{ss}})\hat{n}_t \tag{35}$$

$$\hat{i}_t = \hat{r}_t + E_t\hat{\pi}_{t+1} \tag{36}$$

$$\hat{\lambda}_t = -E_t(\eta_c\hat{m}_{t+1} + \hat{\pi}_{t+1}) \tag{37}$$

$$\hat{m}_t = \hat{m}_{t-1} - \hat{\pi}_t + \mu_t \tag{38}$$

To complete the model, we add the disturbances:

$$\hat{z}_t = \rho_z\hat{z}_{t-1} + \epsilon_{z,t} \tag{39}$$

$$\hat{\mu}_t = \rho_\mu\hat{\mu}_{t-1} + \phi_z\hat{z}_{t-1} + \epsilon_{\mu,t}\ , \quad \phi_z = 0 \tag{40}$$

Dynamics

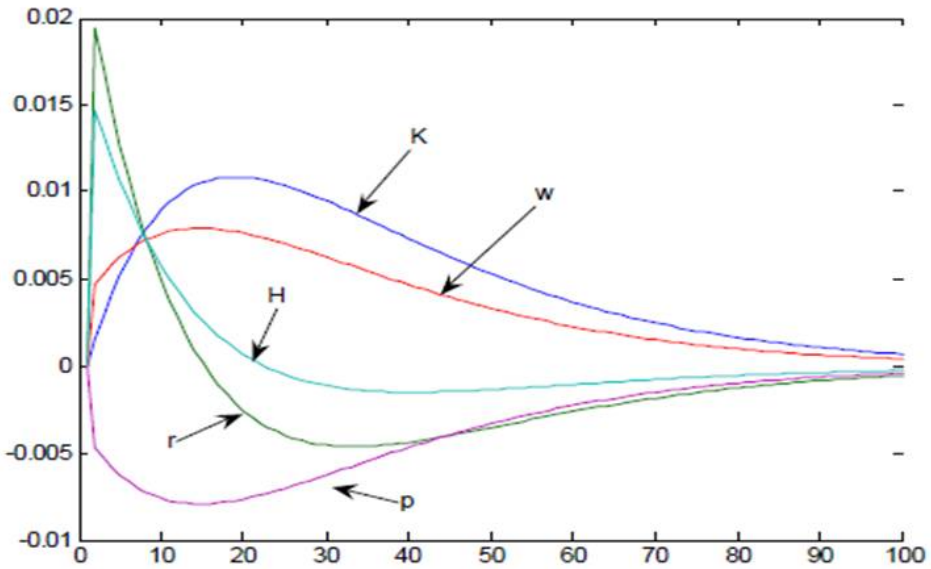
□ Updating (38) one period ahead and substituting into (37), and combining (35), we have:

$$\hat{y}_t = (1 + \eta_n \frac{n^{ss}}{1 - n^{ss}})\hat{n}_t + \eta_c(\hat{m}_t + E_t\mu_{t+1}) - (\eta_c - 1)E_t\hat{\pi}_{t+1} \qquad (41)$$

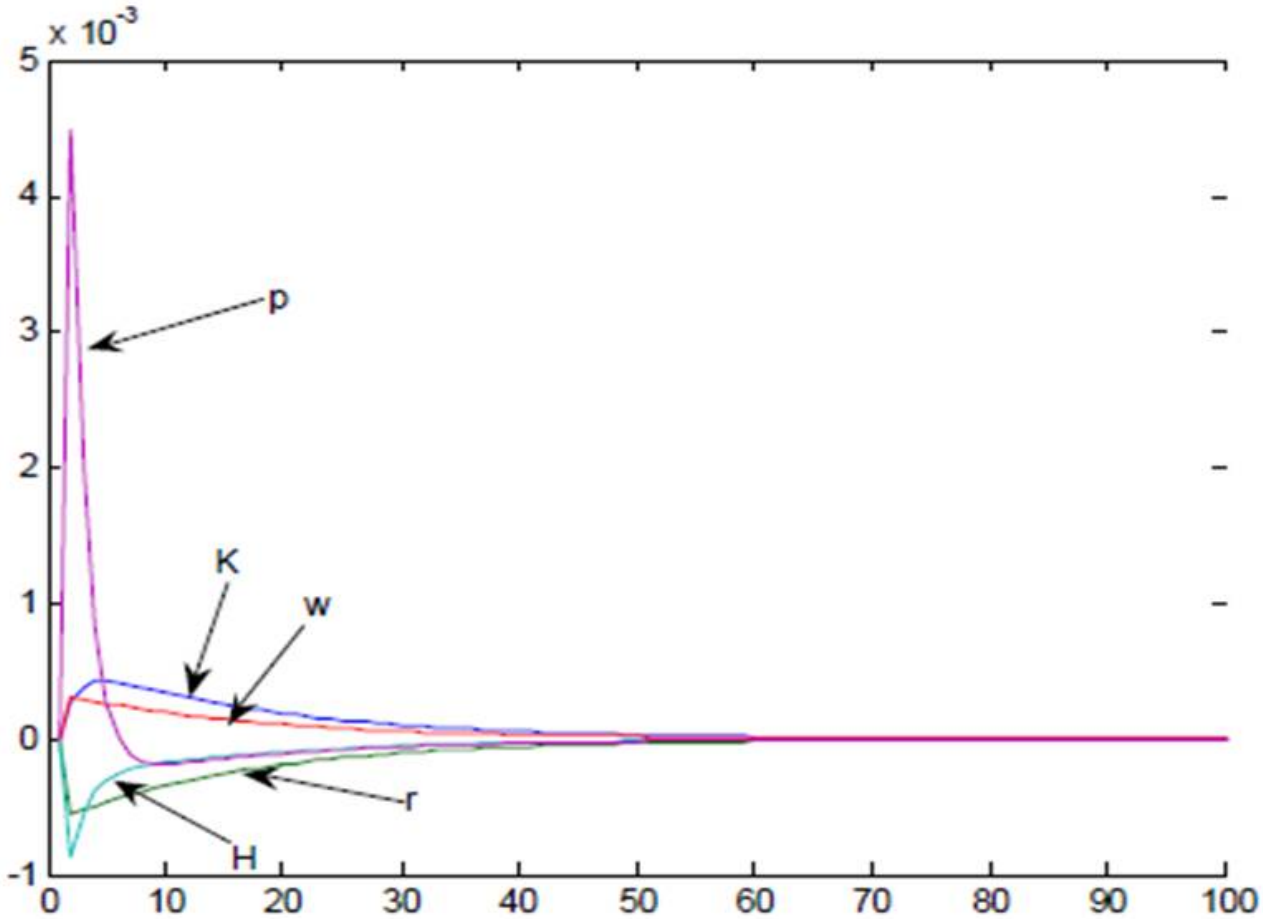
The substitution retains both η_c and expected inflation. For $\eta_c = 1$, it reduces to the familiar expression without the inflation term. With the calibrated $\eta_c = 2$, the full expression is required; the labour response must be determined jointly with the other equilibrium conditions.

□ Now, let's now look at the impulse responses of the model in respect to both the real shock and nominal shock.

Dynamics (Impulse responses to a shock in technology)



Dynamics (Impulse responses to a shock in money)



THE END

Appendix A: First Order Conditions

Value function:

$$\begin{aligned} V(k_{t-1}, b_{t-1}, m_{t-1}, z_t, \mu_t) = \max[u(\frac{c_t^{1-\eta_c}}{1-\eta_c} + \frac{(1-n_t)^{1-\eta_n}}{1-\eta_n}) + \beta E_t V(k_t, m_t) \\ +\lambda_t(\omega_t - c_t - m_t - k_t) \\ +\Lambda_t(\frac{m_{t-1}}{1+\pi_t} + \tau_t - c_t)] \end{aligned} \tag{A.1}$$

where the maximization is with respect to $\{c_t, n_t, k_t, m_t\}$.

FOCs:

- c_t

$$\frac{\partial V(k_{t-1}, b_{t-1}, m_{t-1})}{\partial c_t} = c^{-\eta_c} - \lambda_t - \Lambda_t \tag{A.2}$$

$$\lambda_t + \Lambda_t = c^{-\eta_c} \tag{A.3}$$

Appendix A: First Order Conditions

FOCs:

- n_t[4]

$$\frac{\partial V(k_{t-1}, b_{t-1}, m_{t-1}, z_t, \mu_t)}{\partial n_t} = -(1-n_t)^{-\eta_n} + (1-\alpha) z_t k_{t-1}^{\alpha} n_t^{-\alpha} \lambda_t \quad \text{(A.4)}$$

$$(1-\alpha) z_t k_{t-1}^{\alpha} n_t^{-\alpha} \lambda_t = (1-n_t)^{-\eta_n} \quad \text{(A.5)}$$

- k_t

$$\frac{\partial V(k_{t-1}, b_{t-1}, m_{t-1}, z_t, \mu_t)}{\partial k_t} = \frac{\partial \beta E_t V_k(k_t, m_t)}{\partial k_t} - \lambda_t \quad \text{(A.6)}$$

The associated envelope condition:

$$\frac{\partial V(k_{t-1}, b_{t-1}, m_{t-1}, z_t, \mu_t)}{\partial k_{t-1}} = V_k(k_{t-1}, m_{t-1})[f_k(k_{t-1}, n_t, z_t) + (1-\delta)] \quad \text{(A.7)}$$

[4] $k_{t-1} = f(k_{t-1}, n_t, z_t) + (1-\delta)k_{t-1} + \tau_t + \frac{m_{t-1}}{1+\pi_t}$, where $f(k_{t-1}, n_t, z_t) = z_t k_{t-1}^{\alpha} n_t^{1-\alpha}$

Appendix A: First Order Conditions

FOCs:

- k_t

  Updating (A.7):

$$\frac{\partial V_k(k_t, m_t)}{\partial k_t} = V_k(k_t, m_t)[f_k(k_t, n_{t+1}, z_{t+1}) + (1 - \delta)] \tag{A.8}$$

  where $f_k(k_t, n_{t+1}, z_{t+1}) + (1 - \delta) = R_t$, the gross real rate of return [5] $(1 + r_t)$, which is 1 plus the MP_k net of depreciation.

  Substitute (A.8) into (A.6) and set equal to zero:

$$\beta R_t E_t V_k(k_t, m_t) = \lambda_t \tag{A.9}$$

[5] Note, the real rate of return of capital stock r_t^k is equal to the real rate of interest r_t (no other assets such as bond in the model).

Appendix A: First Order Conditions

FOCs:

- b_t

$$\frac{\partial V(k_{t-1}, b_{t-1}, m_{t-1}, z_t, \mu_t)}{\partial b_t} = -\lambda_t - \Lambda_t + \frac{\partial \beta E_t V_b(k_t, m_t)}{\partial b_t} \tag{A.10}$$

The associated envelope condition [6]:

$$\frac{\partial V(k_{t-1}, b_{t-1}, m_{t-1}, z_t, \mu_t)}{\partial b_{t-1}} = \lambda_t V_b(k_{t-1}, b_{t-1}, m_{t-1})(\frac{1 + i_{t-1}}{1 + \pi_t}) \tag{A.11}$$

Update (A.11) and substitute into (A.10) and set equal to zero:

$$\lambda_t + \Lambda_t = \beta R_t E_t V_b(k_t, m_t)\lambda_{t+1} \tag{A.12}$$

[6] $k_{t-1} = f(k_{t-1}, n_t, z_t) + (1 - \delta)k_{t-1} + \tau_t + \frac{m_{t-1}}{1+\pi_t} + \frac{b_{t-1}(1+i_{t-1})}{1+\pi_t}$; and the CIA constraint: $c_t = \frac{m_{t-1}}{1+\pi_t} + \tau_t - b_t$; that is, BC is as Eq 3.17 and CIA constraint is as 3.20 in the 3rd edition. Take note, later on Walsh changed BC to 3.36. The FOC w.r.t. b_t here is derived based on the assumption that BC (Eq. 5) and CIA constraint (Eq. 6).

Appendix A: First Order Conditions

FOCs:

- m_t

$$\frac{\partial V(k_{t-1}, b_{t-1}, m_{t-1}, z_t, \mu_t)}{\partial m_t} = \frac{\partial \beta E_t V_k(k_t, m_t)}{\partial m_t} - \lambda_t \tag{A.13}$$

The associated envelope condition [7]:

$$\frac{\partial V(k_{t-1}, b_{t-1}, m_{t-1}, z_t, \mu_t)}{\partial m_{t-1}} = \frac{\partial \lambda_t(k_{t-1}, b_{t-1}, m_{t-1}, z_t, \mu_t)}{\partial m_{t-1}} + \frac{\partial \Lambda_t(\frac{m_{t-1}}{1+\pi_t} + \tau_t - c_t)}{\partial m_{t-1}} \tag{A.14}$$

$$= \frac{1}{1+\pi_t}\lambda_t + \frac{1}{1+\pi_t}\Lambda_t \tag{A.15}$$

Update (A.15) and substitute into (A.13):

$$\beta E_t[\frac{\lambda_{t+1} + \Lambda_{t+1}}{1+\pi_{t+1}}] = \lambda_t \tag{A.16}$$

[7] $k_{t-1} = f(k_{t-1}, n_t, z_t) + (1-\delta)k_{t-1} + \tau_t + \frac{m_{t-1}}{1+\pi_t}$

Appendix B: Steady State

- R^{ss}

Recall (16), $\beta R_t E_t \lambda_{t+1} = \lambda_t$, in steady state, $\lambda_{t+1} = \lambda_t = \lambda^{ss}$, therefore we have:

$$R^{ss} = \frac{1}{\beta} \tag{B.1}$$

- $\frac{y^{ss}}{k^{ss}}$

Given the production function, we have:

$$f_k(k_t, n_{t+1}, z_{t+1}) = \frac{\partial y_{t+1}}{\partial k_t} \tag{B.2}$$

$$\frac{\partial y_{t+1}}{\partial k_t} = \alpha z_{t+1} k_t^{\alpha-1} N_{t+1}^{1-\alpha} \tag{B.3}$$

$$\frac{\partial y_{t+1}}{\partial k_t} = \alpha \frac{y_{t+1}}{k_t} \tag{B.4}$$

Appendix B: Steady State

- $\frac{y^{ss}}{k^{ss}}$

substitute (B.4) into $f_k(k_t, n_{t+1}, z_{t+1}) + 1 - \delta = R_t$, we have:

$$R_t = \alpha \frac{y_{t+1}}{k_t} + 1 - \delta \tag{B.5}$$

Substitute (B.1) into (B.5) gives us the steady state output-capital ratio:

$$\frac{1}{\beta} = \alpha \frac{y^{ss}}{k^{ss}} + 1 - \delta \tag{B.6}$$

$$\frac{y^{ss}}{k^{ss}} = \frac{1}{\alpha}\left(\frac{1}{\beta} - 1 + \delta\right) \tag{B.7}$$

$$or \quad \frac{y^{ss}}{k^{ss}} = \frac{1}{\alpha}\left(R^{ss} - 1 + \delta\right) \tag{B.8}$$

Appendix B: Steady State

- $\frac{c^{ss}}{k^{ss}}$

From the resource constraint we know:

$$y_t = c_t + x_t \tag{B.9}$$
$$y_t = c_t + k_t - (1-\delta)k_{t-1} \tag{B.10}$$
$$y_t = c_t + k_t - k_{t-1} + \delta k_{t-1} \tag{B.11}$$

In steady state (B.10) becomes:

$$y^s = c^{ss} + k^{ss} - k^{ss} + \delta k^{ss} \tag{B.12}$$
$$\frac{c^{ss}}{k^{ss}} = \frac{y^{ss}}{k^{ss}} - \delta \tag{B.13}$$

Appendix B: Steady State

- $\frac{n^{ss}}{k^{ss}}$

In steady state the production function $y_t = z_t k_{t-1}^{\alpha} n_t^{1-\alpha}$ becomes:

$$y^{ss} = (k^{ss})^{\alpha}(n^{ss})^{1-\alpha} \tag{B.14}$$

and

$$\frac{y^{ss}}{k^{ss}} = (k^{ss})^{\alpha-1}(n^{ss})^{1-\alpha} \tag{B.15}$$

$$\frac{y^{ss}}{k^{ss}} = (\frac{n^{ss}}{k^{ss}})^{1-\alpha} \tag{B.16}$$

$$\frac{n^{ss}}{k^{ss}} = (\frac{y^{ss}}{k^{ss}})^{\frac{1}{1-\alpha}} \tag{B.17}$$

Appendix B: Steady State

- n^{ss}

Recall the FOC of n_t (A.5):

$$(1-\alpha)z_t k_{t-1}^{\alpha} n_t^{-\alpha} \lambda_t = (1-n_t)^{-\eta_n} \qquad ((A.5))$$

and

$$(1-n_t)^{-\eta_n} = ((1-\alpha)z_t k_{t-1}^{\alpha} n_t^{1-\alpha} \frac{1}{n_t} \lambda_t \qquad (B.18)$$

$$(1-n_t)^{-\eta_n} = (1-\alpha)\frac{y_t}{n_t}\lambda_t \qquad (B.19)$$

In steady state, we have:

$$(1-n^{ss})^{-\eta_n} = (1-\alpha)\frac{y^{ss}}{n^{ss}}\lambda^{ss} \qquad (B.20)$$

$$(1-n^{ss})^{-\eta_n} = (1-\alpha)\frac{y^{ss}}{k^{ss}}(\frac{n^{ss}}{k^{ss}})^{-1}\lambda^{ss} \qquad (B.21)$$

Appendix B: Steady State

- n^{ss}

As we know the steady states of the FOCs of (A.3) and (A.16) are

$$c^{ss} = \lambda^{ss} + \Lambda_{ss} \tag{B.22}$$

$$\lambda_{ss} = \beta\left(\frac{\lambda^{ss} + \Lambda^{ss}}{1 + \pi^{ss}}\right) \tag{B.23}$$

Substitute (B.22) into (B.23), we have:

$$\lambda^{ss} = \beta\frac{(c^{ss})^{-\eta_c}}{1 + \pi^{ss}} \tag{B.24}$$

Appendix B: Steady State

- n^{ss}

Now substitute (B.24) into (B.21) and replace $(c^{ss})^{-\eta_c}$ with $(\frac{c^{ss}}{k^{ss}})^{-\eta_c}(\frac{k^{ss}}{n^{ss}})^{-\eta_c}(n^{ss})^{-\eta_c}$, we have:

$$(1-n^{ss})^{-\eta_n} = (1-\alpha)\frac{\beta}{1+\pi^{ss}}(\frac{y^{ss}}{k^{ss}})(\frac{n^{ss}}{k^{ss}})^{-1}(\frac{c^{ss}}{k^{ss}})^{-\eta_c}(\frac{k^{ss}}{n^{ss}})^{-\eta_c}(n^{ss})^{-\eta_c} \quad \text{(B.25)}$$

$$(1-n^{ss})^{-\eta_n}(n^{ss})^{\eta_c} = (1-\alpha)\frac{\beta}{1+\pi^{ss}}(\frac{y^{ss}}{k^{ss}})(\frac{c^{ss}}{k^{ss}})^{-\eta_c}(\frac{n^{ss}}{k^{ss}})^{\eta_c-1} \quad \text{(B.26)}$$

Substitute (B.17) into (B.26), we have:

$$(1-n^{ss})^{-\eta_n}(n^{ss})^{\eta_c} = (1-\alpha)\frac{\beta}{1+\pi^{ss}}(\frac{y^{ss}}{k^{ss}})(\frac{c^{ss}}{k^{ss}})^{-\eta_c}(\frac{y^{ss}}{k^{ss}})^{\frac{\eta_c-1}{1-\alpha}} \quad \text{(B.27)}$$

$$(1-n^{ss})^{-\eta_n}(n^{ss})^{\eta_c} = (1-\alpha)\frac{\beta}{1+\pi^{ss}}(\frac{c^{ss}}{k^{ss}})^{-\eta_c}(\frac{y^{ss}}{k^{ss}})^{\frac{\eta_c-\alpha}{1-\alpha}} \quad \text{(B.28)}$$

Appendix B: Steady State

- $\pi^{ss} = \theta^{ss}$

Recall the CIA constraint in nominal term:

$$C_t = M_{t-1} + T_t \tag{B.29}$$

in real term:

$$c_t = M_{t-1}/P_t + \tau_t \tag{B.30}$$

Substitute $\tau_t = \frac{M_t - M_{t-1}}{P_t}$ into (B.30), we have:

$$c_t = m_t \tag{B.31}$$

and in steady state:

$$c^{ss} = m^{ss} \tag{B.32}$$